

PEDRO HENRIQUE BOHLING PERES

Mechatronics Engineering Student



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PROFILE

Passionate about Robotics and Computing, Mechatronics Engineering student and researcher, I seek to improve my skills and develop skills and technical knowledge in these areas. I want to be able to assimilate knowledge into projects and participate in the development of new technologies.

SKILLS

- Machine Learning | Deep Learning
- Robotics
- ROS2
- Docker
- C/C++/C#, Python, Java and Dart Programming Languages
- Data Analysis
- Cross-Platform Software Development (Flutter)
- Computer Vision
- Embedded Systems
- Game Development
- System Control

EXPERIENCES

NOV 2023 - PRESENT | EDROM (MOBILE ROBOTICS DEVELOPMENT TEAM)

- Captain (AUG 2025 - PRESENT)

As captain of the Equipe de Desenvolvimento em Robótica Móvel (EDROM), I am responsible for the project as a whole, which means I must organize, plan, and manage everything necessary to make the project happen, whether delegating activities to members and area directors or working on the project myself.

- **Project Director - Movement (DEZ 2024 - PRESENT)**

As Director of the movement project area, I am responsible for leading and organizing activities related to programming, control and hardware necessary to make it possible to move robot joints and subsequently move them. Some of these activities include research and application: Control system and conversion of movements and positions between linear actuators and axial motors, in addition to training a humanoid walking optimization model using reinforcement learning.

- **Movement and Behaviour Member (FEV 2024 - DEZ 2024)**

I participated in the project of two humanoid robots that aimed to play football autonomously in the RoboCup Soccer League competition, for this, I worked in the behavior area with the development of codes for the use of IMUs that went on ESP32 microcontrollers, in addition to integrating everything with ROS2. Regarding movement, I participated in the development of a control system to perform the movement of the legs of one of the robots that had degrees of freedom defined by linear actuators instead of conventional servos.

- **Trainee (NOV 2023 - FEV 2024)**

During the trainee process, I obtained training and carried out activities involving all areas of the project, namely: Vision (Training a neural network for object recognition with Yolo), Electrical (Use of KiCad Software for PCBs), Structure (Simulation of stresses and modeling of parts in SolidWorks), Behavior (Development of Scripts using State Machines), Movement (Calculation of inverse and direct kinematics of a two-joint robotic arm and subsequent application in a script).

JUNE 2025 - MAR 2026 | RESEARCHER - CROB (LABROM) - EESC USP

- **Researcher in the Mobile Robotic Group**

Overall, I worked with robotics and related algorithms to assist in a Petrobras robotics project, implementing algorithms for generating meshes for simulation environments and deploying policies in quadruped robots, such as B1, Spot and ANYmal.

Developed and implemented a Model Predictive Control (MPC) architecture for the Boston Dynamics Spot robot to ensure base stability during complex load manipulation tasks. Transitioned the control system from a reactive Whole-Body Control approach to a predictive model utilizing rigid body dynamics and OSQP solver. Engineered feedforward torque compensation and prediction horizons to proactively counteract disturbances generated by the robotic arm, significantly enhancing the quadruped's dynamic equilibrium and real-time operational robustness.

MAY 2025 | OCT 2025 COMPUTER VISION INTERN - UAILOOK

- **Development of Computer Vision and AI Projects**

MAR 2025 - APRIL 2026 | SCIENTIFIC INITIATION - EDROM | LAR

- Development, Simulation and Optimization of a Walking Model for Humanoid Robots

In general, I work with the development of the mathematical model for a bipedal humanoid robot walk, to be simulated and applied in the physical system, in addition, optimizing techniques will be applied.

OUT 2023 - MAR 2025 | SCIENTIFIC INITIATION - UFU FACOM

- Generation of Databases and use of Large Language Models(LLMs) in Automatic Adaptive Player Systems for Education

My main role is in doctoral research that seeks to use data generated by users of certain games to feed LLMs that would allow intelligent agents to readapt, focusing and prioritizing the user's cognitive weaknesses perceived by the model, so that they can be exercised and improved.

TRAININGS AND COURSES

DEEP LEARNING WITH PYTHON

Artificial, convolutional, recurrent neural networks, self-organizing maps, boltzmann machines, autoencoders and GANs

DATA SCIENTIST TRAINING

FLUTTER FOR ANDROID AND IOS APP DEVELOPMENT

GAME DEVELOPMENT (JAVA, C#, UNITY AND GODOT)

EDUCATION AND CERTIFICATIONS

- UNDERGRADUATE IN MECHATRONICS ENGINEERING (2023 - PRESENT)
- INDUSTRIAL AUTOMATION TECHNICIAN (NOV 2021 - ABR 2023)
- OBA - BRAZILIAN ASTRONOMY OLYMPICS MEDALHIST (GOLD - 2012, 2013, 2019 / SILVER - 2022)
- APROVA+ (MATHEMATICS SIXTH POSITION)
- OBMEP - BRAZILIAN MATHEMATICS OLYMPIAD FOR PRIVATE SCHOOLS (HONORED MENTION - 2018)
- MATH KANGAROO (GOLD - 2022)
- NATIONAL SCIENCE OLYMPIAD (HONORED MENTION - 2022)
- MATHÉMATIQUES SANS FRONTIÈRES NATIONAL (GOLD - 2018 | SILVER - 2022 | BRONZE - 2019)
- MATHÉMATIQUES SANS FRONTIÈRES REGIONAL (GOLD - 2018, 2022 | SILVER - 2019)
- GOIANA ASTRONOMY OLYMPICS (GOLD - 2018, 2021, 2022)
- GOIANA ENVIRONMETAL OLYMPICS (GOLD - 2021, 2022)
- 3RD PLACE IN HUMANOID SOCCER LEAGUE AT CBR 2025 (COMPETIÇÃO BRASILEIRA DE ROBÓTICA)

EXTRACURRICULAR ACTIVITIES

- ALGORITHMS AND COMPUTER PROGRAMMING TEACHER'S ASSISTANT
- COMPETITOR AT ROBÓTICA 2024 (CBR 2024)
- COMPETITOR AT ROBÓTICA 2025 (CBR 2025)
- SEMATRON 2024
- SEMATRON 2025
- COMPETITOR AT XXI TRIÂNGULO MINEIRO REGIONAL PROGRAMMING MARATHON